Cheat sheet

Maths & Statistics

Linear algebra, probability, statistics and calculus for AI

Linear algebra

Vector / matrix / tensor	1-D list / 2-D grid / n-D array of numbers
Dot product	$a \cdot b = \Sigma\, a_i b_i$; large when vectors align
Cosine similarity	$a \cdot b \,/\, (\|a\|\|b\|)$ in [−1, 1]
Matrix multiply	(m×n)(n×p) → (m×p); each layer of a network is $XW + b$
Eigenvectors / PCA	Directions of maximum variance
Norms	L1 = $\Sigma\|x\|$ (sparsity), L2 = $\sqrt{\Sigma x^2}$ (regularisation)

Probability

Conditional	$P(A\|B) = P(A \cap B) \,/\, P(B)$
Bayes	$P(A\|B) = P(B\|A)\, P(A) \,/\, P(B)$
Independence	$P(A \cap B) = P(A)\, P(B)$
Expected value	$E[X] = \Sigma\, x\, P(x)$
Distributions	Bernoulli, binomial, Poisson (counts), normal, uniform, exponential
Softmax	$\exp(z_i) \,/\, \Sigma \exp(z_j)$; temperature divides z before softmax

Statistics

Central tendency	Mean (sensitive to outliers), median (robust), mode
Spread	Variance, standard deviation, IQR
CLT	Sample means are approximately normal for large n
Hypothesis test	H_0 vs H_1; p-value < α (e.g. 0.05) → reject H_0

Errors	Type I = false positive (α); Type II = false negative (β); power = 1 − β
Confidence interval	Range likely to contain the true parameter at a given confidence
Tests	t-test (means), chi-square (categories), ANOVA (3+ groups)
Correlation	Pearson (linear), Spearman (rank); correlation ≠ causation

Calculus and optimisation

Derivative	Rate of change / slope
Chain rule	`d/dx f(g(x)) = f'(g(x)) g'(x)`, the core of backprop
Gradient descent	`θ ← θ - η ∇L(θ)`
Convex	One global minimum; neural nets are non-convex

Information theory

Entropy	Uncertainty: `-Σ p log p`
Cross-entropy	Loss for classification / next-token prediction
KL divergence	How one distribution differs from another (used in RLHF, distillation)
Perplexity	`exp(cross-entropy)`; lower = better language model